

Resilient Edge Ecosystem for Collaborative and Trustworthy Disaster Response (REsCue): An Information-Centric Approach

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What is NSF RINGS?

- Resilient & Intelligent NextG Systems (RINGS)
- Collaboration across academia, government, and industrial partners
 - Department of Defense (DoD)
 - Office of the Under Secretary of Defense for Research and Engineering
 - National Institute of Standards and Technology (NIST)
 - Industry Partners: Apple, Ericsson, Google, IBM, Intel, Microsoft, Nokia, Qualcomm



NIST



ERICSSON
Google

IBM

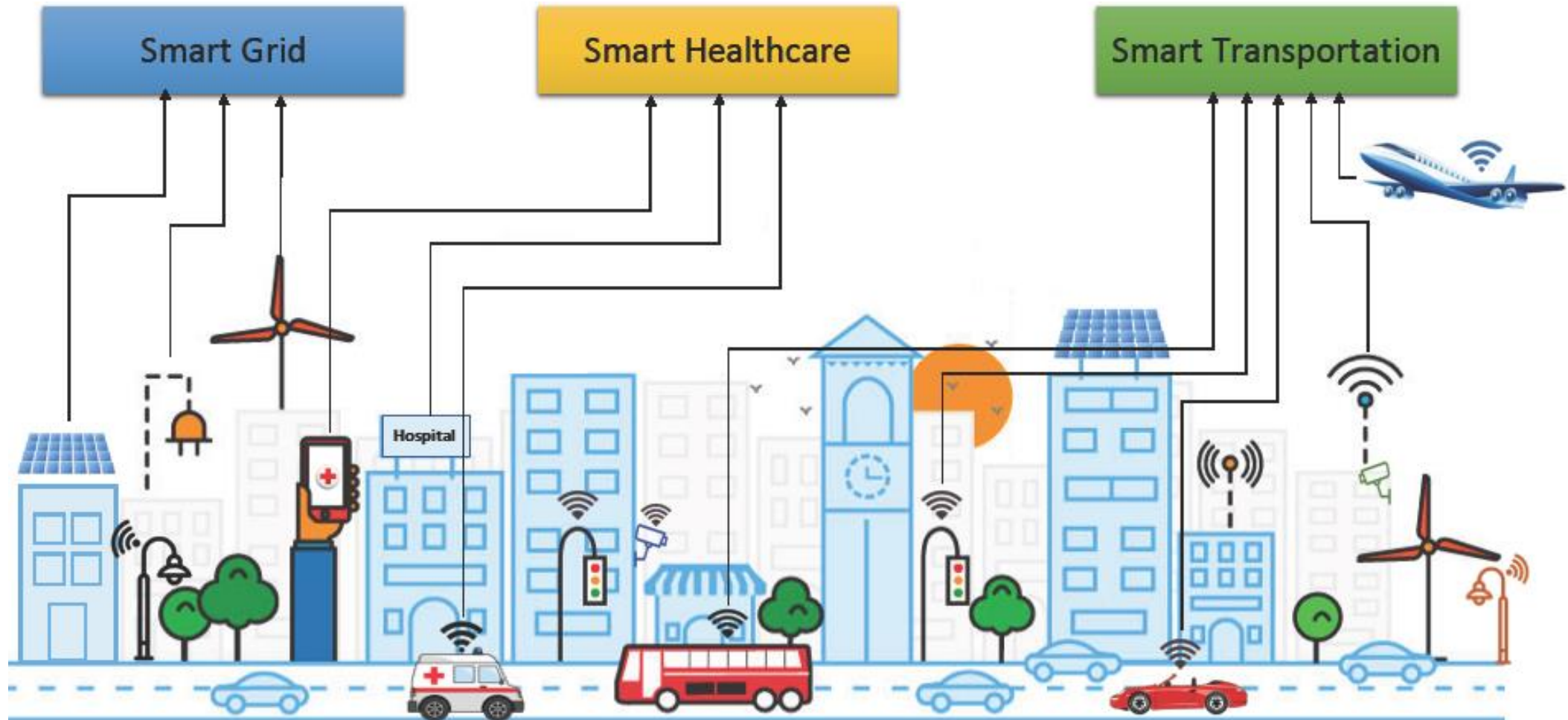
intel.



NOKIA

Qualcomm
Qualcomm
Technologies, Inc.

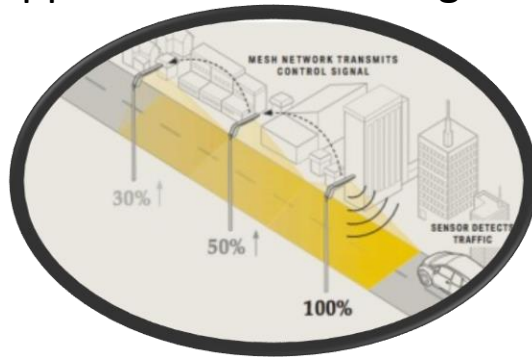
Smart-city is essentially a network of autonomous networks.



Why Do We Need **Yet** Another Architecture?

Existing Smart City Architectures

Application driven designs



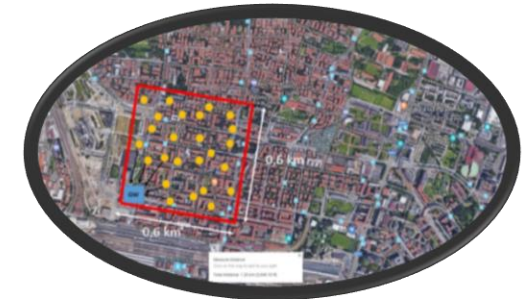
*MDPI Sensors Journal 2018

Novel communication technologies



*IEEE Internet of Things Journal 2018

Real world deployments



*Smart City Pilot Projects Using LoRa and IEEE802.15.4 Technologies

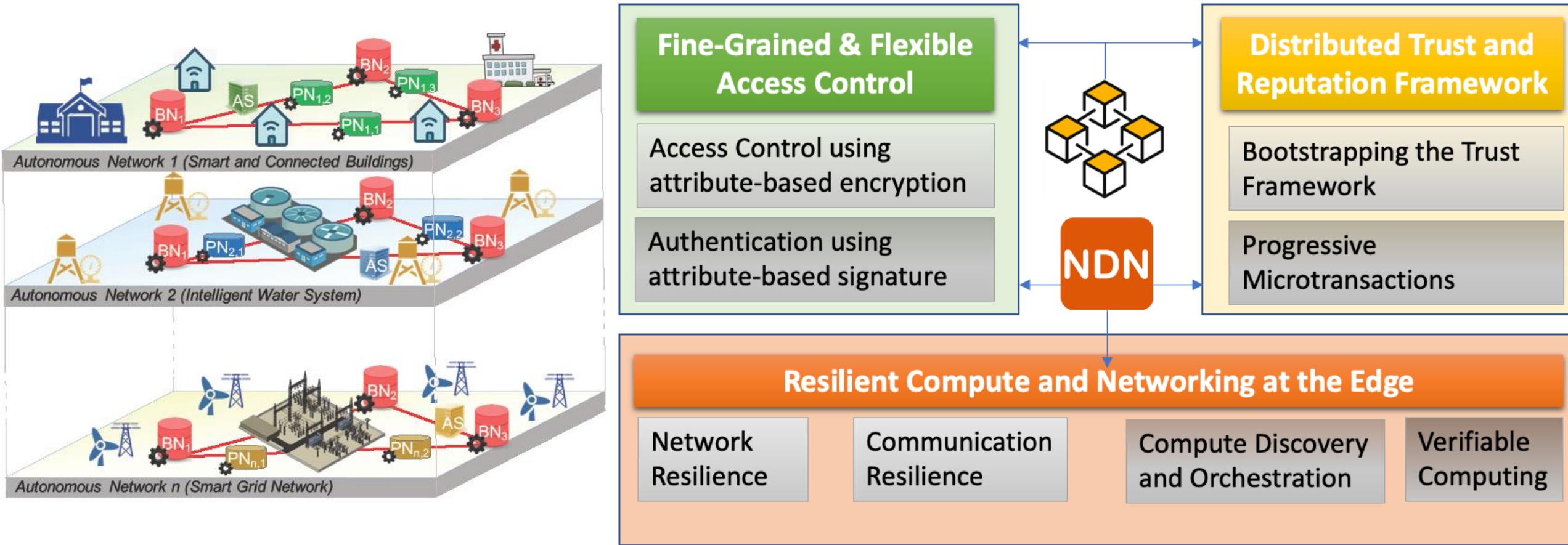
=> Push tasks to the network layer architecture to reduce complexity

- Better Scalability; Reusability; Efficiency; Security

Outline

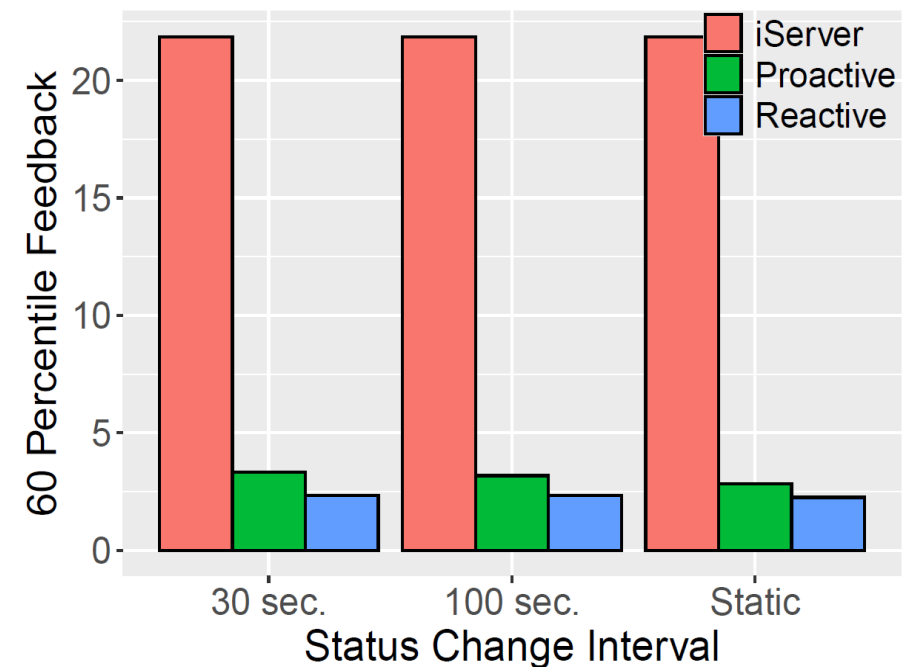
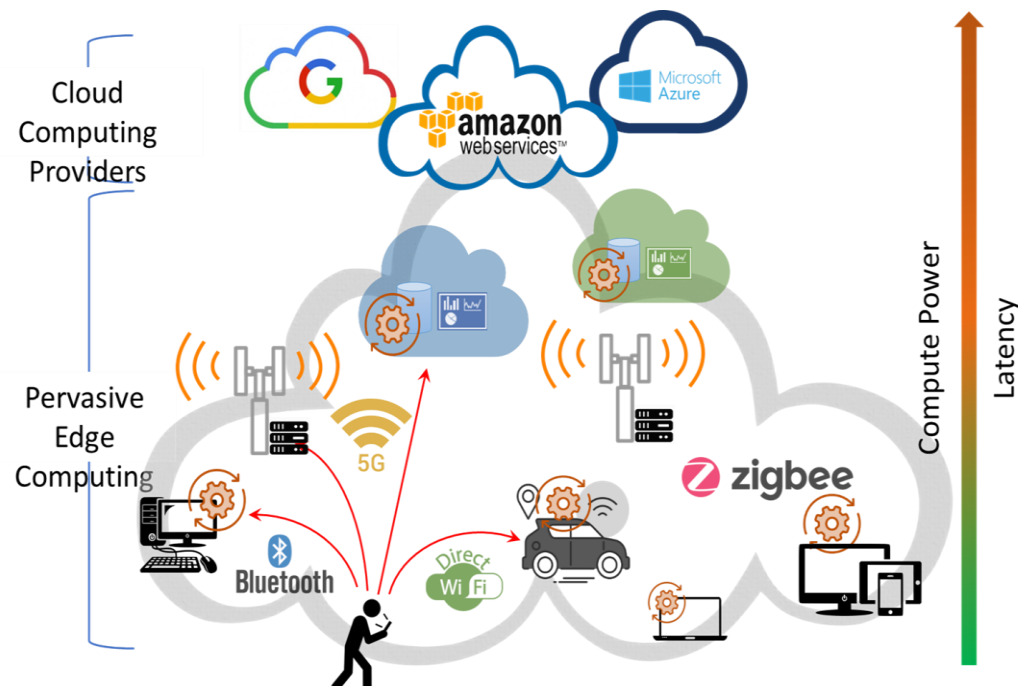
- Motivation
 - Existing Architecture
 - What are we lacking?
- An Information Centric Architecture
- Data and Network Resiliency
- Information Centric Security
- Conclusion

ResCue: An Information Centric Architecture



Edge Computing Architecture (Service Discovery, Load Balancing)

- Need for a seamless, self organizing, and resilient edge computing ecosystem
 - A Pervasive Edge Computing (PEC) which can be formed in an ad-hoc fashion
 - User devices can serve as compute devices (join and leave the network as needed)
 - A distributed networking architecture with minimum to no centralized services (DNS, addressing, etc.)



G. Torres *et al.*, "uDiscover: User-Driven Service Discovery in Pervasive Edge Computing using NDN," 2022 IEEE International Conference on Edge Computing and Communications (EDGE), 2022, pp. 77-82, doi: 10.1109/EDGE55608.2022.00022.

Resilience (Network and Wireless)

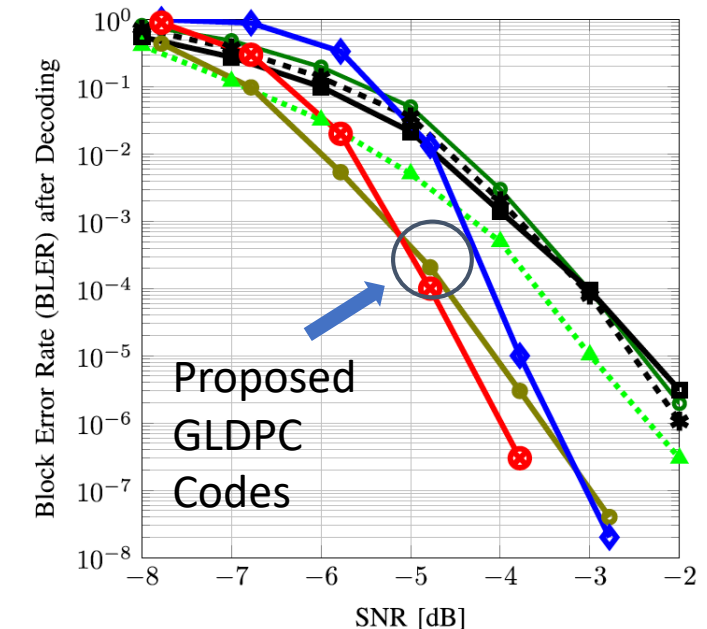
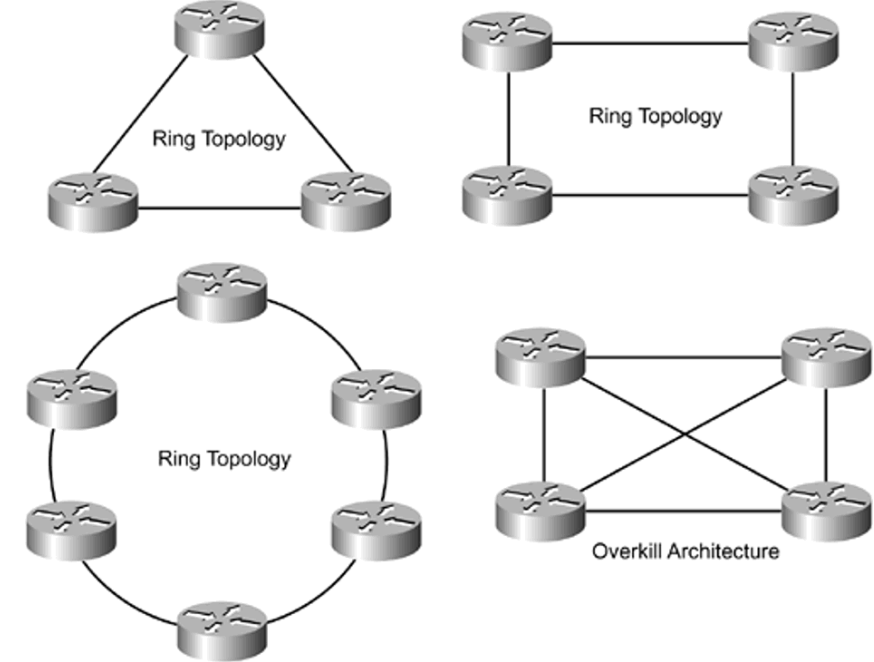
Goal: Explore survivable network design for network resilience and advanced channel coding techniques to address communication resilience with 5G URLLC.

Technology Used:

1. Pre-deployed survivable network design configurations
2. Improved error control performance for very short block lengths
3. Low graph (algorithmic/implementation) complexity

Facets in evaluation:

1. Network tolerance to disconnections/dirruptions.
2. Performance under limited/no HARQ?
3. Pursue favorable tradeoffs with BER/spectral efficiency
4. Theoretical performance guarantees under latency and power consumption constraints

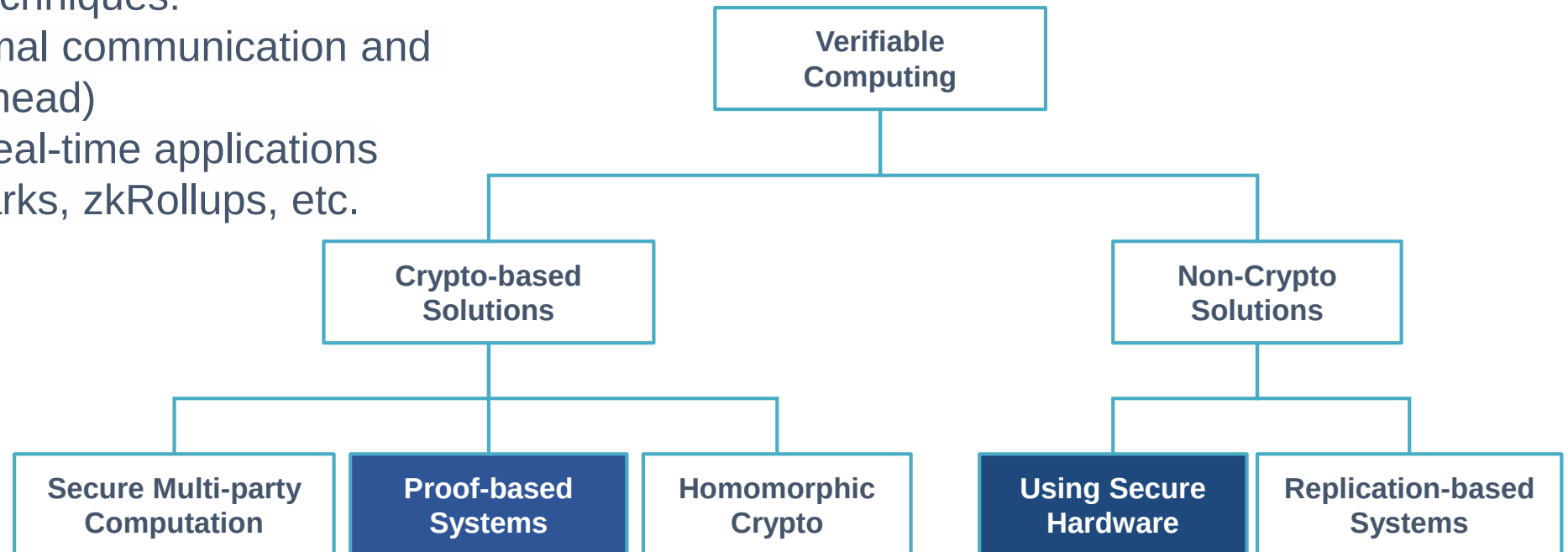


Verifiable Computing using SGX and crypto.

Goal: Cloud computing platforms are inherently more trustworthy compared to edge computing platforms due to their centralized nature^[1]. Verification of the offloaded services is required for critical edge applications.

Benefits of selected techniques:

1. Lightweight (minimal communication and computation overhead)
2. Compatible with real-time applications
3. Leveraging zkSnarks, zkRollups, etc.



Trust and Reputation: Leveraging localized distributed ledger and microtransactions to bootstrap trust.

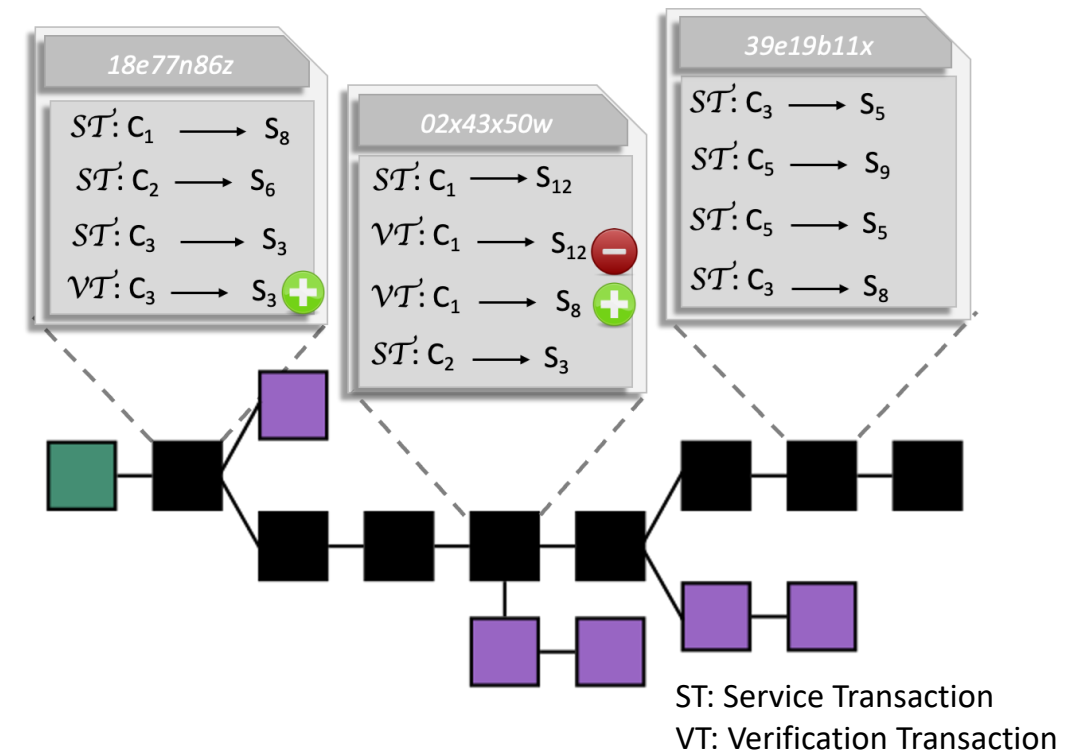
Goal: Modeling trust as a fuzzy truth value build upon entities' reputation over time for decentralized ecosystems.

Benefits of leveraging distributed ledger:

1. Automation and decentralized arbitration through smart contracts
2. Maintaining immutable (micro-)transactions (service executions history)
3. Fair payment and auditability

Facets in adjusting reputation score and trust degree:

1. Service output verification (verifiable computing)
2. Meeting the expected QoS requirements
3. Availability (e.g., service and resource)



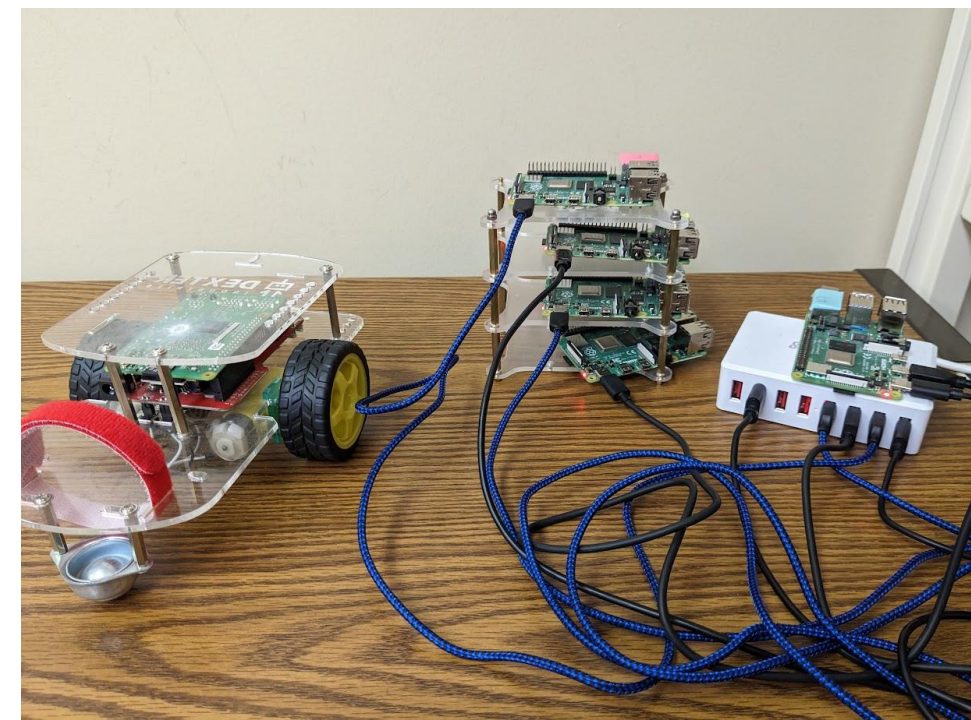
Testbed

Today's testbeds are hard to configure and difficult to setup

- Often designed for a specific app
- “Limited”
- Takes time to setup/run

Build a generic, configurable, and “easy to use”
Testbed:

- Seamless topology settings
- NDN/IP modes
- Wireless/wired modes
- Networking and edge computing scenarios



Thanks!

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